

AC Repair & Service Training

10 Days Professional Training Program

THEORY + PRACTICAL

DAY 1

Basic Electrical & Safety

Build a solid foundation in electrical theory and safe working practices before touching any AC equipment. Safety is the first priority in every professional technician's career.

Electrical Safety Rules

Lockout/tagout procedures, PPE usage, and safe working distances around live circuits.

AC & DC Basics

Understanding alternating and direct current, waveforms, frequency, and their roles in AC systems.

Voltage, Current & Resistance

Ohm's Law in practice — measuring and calculating electrical parameters in real circuits.

Multimeter & Components

Introduction to digital multimeter usage and identification of basic electrical components.

Tools Handling & Testing Instruments

Hands-on familiarization with every essential tool a professional AC technician uses daily. Proper tool handling ensures accuracy, efficiency, and safety on every job site.

1

Digital Multimeter

Measure voltage, current, resistance, and continuity across AC electrical circuits with precision.



Manifold Gauge

Read high and low-side refrigerant pressures to diagnose charge levels and system performance.



Copper Tube Cutter

Clean, burr-free cuts on copper refrigerant lines for leak-proof flare and swage connections.



Clamp Meter

Non-contact current measurement for safe diagnostics on live conductors and compressor loads.



Vacuum Pump

Evacuate moisture and non-condensables from the refrigerant circuit before gas charging.



Flaring & Swaging Tools

Create professional-grade copper joint connections for secure, long-lasting refrigerant piping.

Air Conditioner Basics

Understand the different categories of air conditioning systems and learn to identify every major component inside and out. This knowledge underpins all diagnostic and repair work ahead.

1

Window AC

All-in-one single-unit design installed in a window frame. Ideal for single rooms. Self-contained refrigerant circuit, integrated controls, and straightforward servicing.

2

Split AC

Separate indoor evaporator and outdoor condenser/compressor units. Most common residential and commercial installation. Quieter operation, flexible placement, higher efficiency.

3

Inverter AC

Variable-speed compressor technology that modulates capacity to match cooling demand. Up to **60% more energy-efficient** than fixed-speed units. Requires specialized diagnostic skills.

The Refrigeration Cycle

Master the core science behind every air conditioning system. Understanding how heat is moved — not created or destroyed — is the key to diagnosing any fault correctly.

01

Compressor

Compresses low-pressure refrigerant vapor into high-pressure, high-temperature gas. The heart of the system.

02

Condenser

Releases heat from the hot refrigerant gas to the outside air, converting it back to a high-pressure liquid.

03

Expansion Device

Drops the refrigerant pressure rapidly, causing it to cool dramatically before entering the evaporator.

04

Evaporator

Absorbs indoor heat into the cold refrigerant, cooling the air blown across it into the room.

Refrigerant Types

R22 (Freon)

Legacy refrigerant. Phased out due to ozone depletion. Still found in older systems requiring servicing.

R410A (Puron)

High-efficiency, zero ozone depletion. Standard in non-inverter split systems. Operates at higher pressures.

R32

Next-generation refrigerant with lower global warming potential. Used in modern inverter AC units.

Electrical Components Testing

Learn systematic testing procedures for every major electrical component in an AC system. Accurate component testing is what separates a professional technician from a parts-swapper.

→ Capacitor Testing

Measure capacitance values with a multimeter. Identify bulged, leaking, or out-of-tolerance run and start capacitors that cause compressor and fan motor failures.

→ Compressor Testing

Check winding resistance between terminals (C, S, R), verify insulation to ground, and test for open or shorted windings using a megohmmeter.

→ Fan Motor Testing

Test indoor blower and outdoor condenser fan motor windings, bearing condition, and starting capacitor to diagnose no-airflow and overheating faults.

→ Contactor & Thermostat/Sensors

Inspect contactor contacts for pitting and burning, test coil voltage, and verify NTC/PTC sensor resistance values against manufacturer temperature-resistance charts.

Installation Training

Proper installation is the foundation of a long-lasting, efficient air conditioning system. A poorly installed AC will never perform to its rated capacity regardless of how good the unit is.



Indoor Unit Installation

Selecting the optimal mounting height and location, wall bracket leveling, securing the backplate, and routing the refrigerant and drain lines through the wall sleeve.



Outdoor Unit Installation

Positioning for adequate airflow clearance, anti-vibration pad placement, pipe routing with correct bends, and securing the unit against wind and seismic movement.



Copper Pipe & Drain Connection

Cutting, bending, flaring, and insulating refrigerant lines. Correct drain pipe slope (1:100 minimum) to prevent condensate backup and water leakage complaints.



Electrical Wiring

Reading wiring diagrams, connecting power supply and inter-unit communication cables, correct cable sizing, earthing, and MCB/isolator installation per safety codes.



DAY 7

Gas Charging & Leak Testing

Correct refrigerant handling is a critical professional skill — and a legal requirement. Learn the complete process from evacuation to verified charge, with zero refrigerant waste.

1

Vacuum Process

Connect vacuum pump via manifold gauge. Pull system down to 500 microns or below to remove all moisture and air before charging.

2

Leak Detection

Electronic leak detector scan of all joints and flare connections. Nitrogen pressure test to 300–400 psi to confirm system integrity before introducing refrigerant.

3

Gas Charging

Charge by weight using refrigerant scale or by superheat/subcooling method. Correct charge is critical — even 10% under-charge reduces efficiency by 20%.

4

Refrigerant Recovery

Safe recovery of refrigerant into approved cylinders before any major repair. Required by environmental law in most countries — never vent to atmosphere.

Fault Finding, Repair & Complete AC Service

Apply everything learned to diagnose real faults and perform a full professional service. Systematic fault-finding and thorough cleaning are what keep customers coming back.

AC Fault Finding & Repair

No Cooling

Check refrigerant charge, compressor operation, capacitor, airflow restriction, and dirty coils.

Water Leakage

Inspect drain pan, drain pipe slope, blocked drain, dirty evaporator coil causing ice formation.

Compressor Not Starting

Test capacitor, contactor, overload protector, compressor windings, and supply voltage.

Fan Problems & Error Codes

Motor winding test, capacitor check, blade obstruction, and manufacturer error code diagnosis via remote or PCB LED flash patterns.

Complete AC Service Procedure

1 Indoor Unit Cleaning

Remove and wash filters, clean evaporator coil with foaming coil cleaner, wipe blower wheel and housing.

2 Outdoor Unit Cleaning

Flush condenser coil fins with water, clear debris from fan area, check refrigerant connections.

3 Performance Testing

Measure inlet/outlet temperature differential (target 8–12°C ΔT), check running amperage, and verify operating pressures are within spec before handing back to customer.

DAY 10

Business Development & Final Assessment

Turn your technical skills into a thriving service business. The final day covers everything you need to launch professionally, handle customers with confidence, and earn your certification.

Service Business Setup

Registering your business, sourcing tools and spare parts, building a supplier network, and creating your first service kit inventory.

Customer Handling

Professional communication, setting expectations, handling complaints, building repeat customers, and gathering referrals for organic business growth.

Service Charge Calculation

Pricing labour, parts, gas, and travel correctly. Creating service invoices, understanding profit margins, and competitive market rate benchmarking.

Assessment & Certificate

Practical skills assessment covering installation, fault-finding, and gas charging. Written theory test. Certificate of completion awarded to all passing candidates.